

Algebra 1
Unit 8
Lesson 7 Practice

Name **Answer Key** _____
Date _____
Period _____

1. For the expression, the first step to completing the square is shown.

$$x^2 + 18x + 56$$

$$(x^2 + 18x) + 56$$

1. Finish factoring the perfect square trinomial in vertex form.

$$(x^2 + 18x + 81) + 56 - 81$$

$$(x + 9)^2 - 25$$

In the expression, the value that was added to create the perfect square trinomial is being multiplied by 1, so the value that should be subtracted from 56 to maintain equivalency is 81.

Complete the table to complete the square without using algebra tiles.

	Process	Equation
	Original equation	$y = -5x^2 - 20x + 60$
2.	Factor out the common factor of the first two terms.	$y = -5(x^2 + 4x) + 60$
3.	Determine the value that makes a perfect square trinomial in the parentheses.	$y = -5(x^2 + 4x + 4) + 60$
4.	Then, to maintain equivalence, multiply the factor by the value. Add the opposite of that value.	$y = -5(x^2 + 4x + 4) + 60 + 20$
5.	Rewrite the perfect square trinomial represented in the parentheses as a binomial squared.	$y = -5(x + 2)^2 + 60 + 20$
6.	Combine like terms.	$y = -5(x + 2)^2 + 80$

Rewrite each equation in vertex form by completing the square. Then, give the key features revealed by both equations.

7. $y = x^2 - 2x - 15$
 $y = (x^2 - 2x + 1) - 15 - 1$
 $y = (x - 1)^2 - 16$

8. Key Features

Key Features					
x-intercept		vertex	$(1, -16)$	domain	All Real #'s
x-intercept		increasing	$x > 1$	range	$y \geq -16$
y-intercept	$(0, -15)$	decreasing	$x < 1$	opens	Up
left end behavior		$x \rightarrow -\infty, y \rightarrow \underline{\infty}$			
right end behavior		$x \rightarrow \infty, y \rightarrow \underline{\infty}$			

9. $y = 2x^2 - 8x + 6$
 $y = 2(x^2 - 4x + 4) + 6 - 8$
 $y = 2(x - 2)^2 - 2$

10. Key Features

Key Features					
x-intercept		vertex	$(2, -2)$	domain	All Real #'s
x-intercept		increasing	$x > 2$	range	$y \geq -2$
y-intercept	$(0, 6)$	decreasing	$x < 2$	opens	Up
left end behavior		$x \rightarrow -\infty, y \rightarrow \underline{\infty}$			
right end behavior		$x \rightarrow \infty, y \rightarrow \underline{\infty}$			